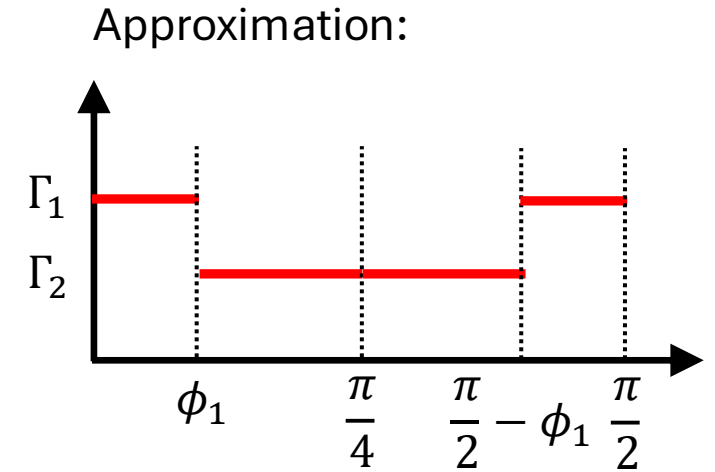
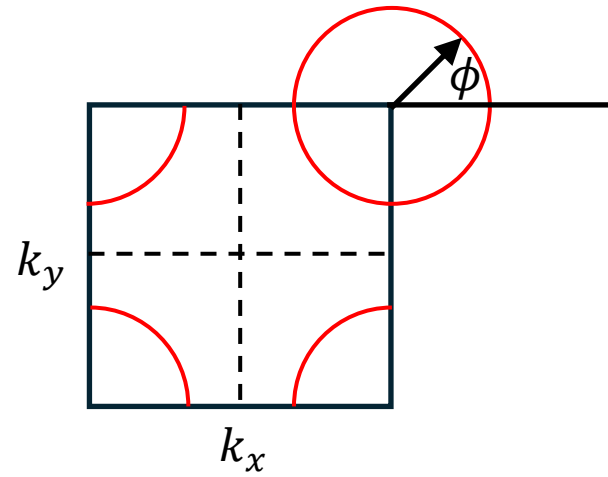


Boltzmann eqs. from PRB (2017)

$$\sigma_{xx} = \frac{e^2}{\pi^3 \hbar} \frac{2\pi}{d} \int d\phi \frac{k_F v_F \cos^2(\phi)}{\Gamma(\phi)},$$

$$\sigma_{xy} = \frac{e^3 H}{\pi^3 \hbar^2} \frac{2\pi}{d} \int d\phi \frac{v_F \cos(\phi)}{\Gamma(\phi)} \frac{\partial}{\partial \phi} \frac{v_F \sin(\phi)}{\Gamma(\phi)},$$



$$\sigma_{xx} \sim \phi_1 \left[\frac{1}{\Gamma_1} - \frac{1}{\Gamma_2} \right] + \frac{1}{4} \frac{\pi}{\Gamma_2}$$

$$\sigma_{xy} \sim \phi_1 \left[\frac{1}{\Gamma_1^2} - \frac{1}{\Gamma_2^2} \right] + \frac{1}{4} \frac{\pi}{\Gamma_2^2}$$

$$\rho_{xx} = \frac{1}{\sigma_{xx}} \sim \frac{\Gamma_1 \Gamma_2}{\phi_1 (\Gamma_2 - \Gamma_1) + \frac{\pi}{4} \Gamma_1}$$

$$\cot \Theta_H = \frac{\sigma_{xx}}{\sigma_{xy}} \sim \frac{\phi_1 (\Gamma_2 - \Gamma_1) + \frac{\pi}{4} \Gamma_1}{\phi_1 \left(\frac{\Gamma_2}{\Gamma_1} - \frac{\Gamma_1}{\Gamma_2} \right) + \frac{\pi}{4} \frac{\Gamma_1}{\Gamma_2}}$$

$$R_H = \frac{\sigma_{xy}}{B \sigma_{xx}^2} \sim \frac{\phi_1 [\Gamma_2^2 - \Gamma_1^2] + \frac{\pi}{4} \Gamma_1^2}{\left[\phi_1 [\Gamma_2 - \Gamma_1] + \frac{\pi}{4} \Gamma_1 \right]^2}$$



similar to transport equations from two bands

Can we obtain $\rho_{xx} \sim T$ and $\cot \theta_H \sim T^2$ from $\Gamma_1(T)$ and $\Gamma_2(T)$?

Set $\Gamma_1 \sim T^{\alpha_1}$, $\Gamma_2 \sim T^{\alpha_2}$ and define $\alpha_{<} = \min(\alpha_i)$, $\alpha_{>} = \max(\alpha_i)$

Large temperature limits:

$$\lim_{T \rightarrow \infty} \rho_{xx} = \lim_{T \rightarrow \infty} \frac{\Gamma_1 \Gamma_2}{\phi_1(\Gamma_2 - \Gamma_1) + \frac{\pi}{4} \Gamma_1} \sim T^{\alpha_{<}}$$

$$\lim_{T \rightarrow \infty} \cot \Theta_H = \lim_{T \rightarrow \infty} \frac{\phi_1(\Gamma_2 - \Gamma_1) + \frac{\pi}{4} \Gamma_1}{\phi_1\left(\frac{\Gamma_2}{\Gamma_1} - \frac{\Gamma_1}{\Gamma_2}\right) + \frac{\pi}{4} \frac{\Gamma_1}{\Gamma_2}} \sim T^{\alpha_{<}}$$



both approach the same temperature dependence

Can we obtain $\rho_{xx} \sim T$ and $\cot \theta_H \sim T^2$ from a temperature dependent crossover angle $\phi_1(T)$?

Set $\Gamma_1 \sim T^{\alpha_1}, \Gamma_2 \sim T^{\alpha_2}, \phi_1 \sim T^\beta$ (of course $\phi_1 < \pi/4$).

Large temperature limits:

$$\lim_{T \rightarrow \infty} \rho_{xx} = \lim_{T \rightarrow \infty} \frac{\Gamma_1 \Gamma_2}{\phi_1 (\Gamma_2 - \Gamma_1) + \frac{\pi}{4} \Gamma_1} \sim T^{\alpha_< - \beta}$$

$$\lim_{T \rightarrow \infty} \cot \Theta_H = \lim_{T \rightarrow \infty} \frac{\phi_1 (\Gamma_2 - \Gamma_1) + \frac{\pi}{4} \Gamma_1}{\phi_1 \left(\frac{\Gamma_2}{\Gamma_1} - \frac{\Gamma_1}{\Gamma_2} \right) + \frac{\pi}{4} \frac{\Gamma_1}{\Gamma_2}} \sim T^{\alpha_<}$$

➡ $\alpha_< = 2$ and $\beta=1$ lead to $\rho_{xx} \sim T$ and $\cot \theta_H \sim T^2$

Example: $\Gamma_1 = 0.002 T^2, \Gamma_2 = 0.2 T^2, \phi_1 = \frac{\pi}{4} \tanh \frac{T}{500}$

